Screen Sketches

By: 3_swarna_1

Tu Reh: 25% contribution

Haris Khan: 25% contribution

Ping Wu: 25% contribution

Bae Meh: 25% contribution

Actors (Haris)

1. Users
    - Interacts with the app to search, listen and rate tunes.

    - Be able to Create an account ,log in and view their rating history.
2. Admins
    - Manages app's content , including overseeing user activity, handling reported issues and managing tune data.
3. Content Moderator
    - Review User Submissions of flagged content.
    - Resolve Disputes by handling user reports.
4. Social media API
    - Share tunes or rating on social media platforms.

Non Functional Requirements (Haris)

1. The app should load pages in less than 3 seconds on average.
2. The app should handle multiple different users without significant performance degradation.
3. The system should support multiple reads and writes per second, ensuring that user reviews, ratings, and album metadata are processed in real time.
4. The app database should be growing constantly to account for new music being released
5. Low downtime while changes are being made
6. We will promptly find and address any bugs that come up

Tables (Haris)

- We will get our data from the Deezer api

User

- User id - primary key
- User name
- User email

- User username
- User password
- User profile pic

Friend

- Friend id - primary key
- User id

Music

- Album id - primary key
- Song id - primary key
- Album genre
- Album artist
- Album title
- Song title
- Album release date
- Album cover
- Album rating

Post

- Post id - primary key
- User id - foreign key
- Album id - foreign key
- Post contents
- Post album rating
- Post date

Message

- User id - foreign key
- Friend id - foreign key
- Message id - primary key
- Message contents
- Message date

ScreenFlow - Developer: Tu Reh

The flowchart for our application clearly outlines the user journey and admin functionalities. It starts with the user deciding to either log in or sign up. If they log in, they are taken to the Home Page,

which is the main hub of the app. From there, users can update their profile, manage friends or followers, chat with friends, share and upload music, and rate tracks they've listened to. There's also an option to log out directly from the Home Page.

User Table:

- **Purpose:** Stores essential user details such as IDs, usernames, emails, passwords, and profile pictures.
- **Functionality:** Used for login authentication, profile management, and identifying users in other features (e.g., posts or chats).

User Authentication and Authorization:

- **Log In/Sign-Up:** Users can securely create accounts and log in using their email and password. This data is validated and stored in the database. Passwords are encrypted for security.
- **Admin Access:** Admins log in through a separate portal. Their elevated privileges are checked through role-based authentication.

New users follow a straightforward sign-up process, creating an account with their email and password. The user information will be stored in the user's table in the database. Where they will be able to access their account through login, after creation.

Chat Table:

- **Purpose:** Logs one-on-one chat messages between friends.
- **Functionality:** Facilitates real-time communication through WebSocket protocols.

Friend Table:

- **Purpose:** Tracks friendships between users.
- **Functionality:** Links users to their friends, allowing them to manage friend lists and initiate chats.

Admin Table:

- **Purpose:** Maintains admin credentials for secure access.
- **Functionality:** Validates admin login and enables access to moderation tools.

Admin Controls:

- Admins oversee content moderation, removing inappropriate material, banning users who violate guidelines, and managing flagged posts or reports.

The flowchart also includes an admin section with a separate login. Admins have specific privileges such as removing inappropriate content, deleting user accounts, and banning users who violate the rules. Admins cannot create an account, it is given to users.

By mapping out the different paths for users and admins, the flowchart provides a clear and organized view of the app's structure and features.

Songs Table:

- **Purpose:** Stores individual song details, linked to albums.
- **Functionality:** Supports detailed exploration of tracks within albums and their respective ratings.

Post Table:

- **Purpose:** Stores user-generated content, including album reviews, posts, and ratings.
- **Functionality:** Allows users to share their music experiences and rate tracks publicly.

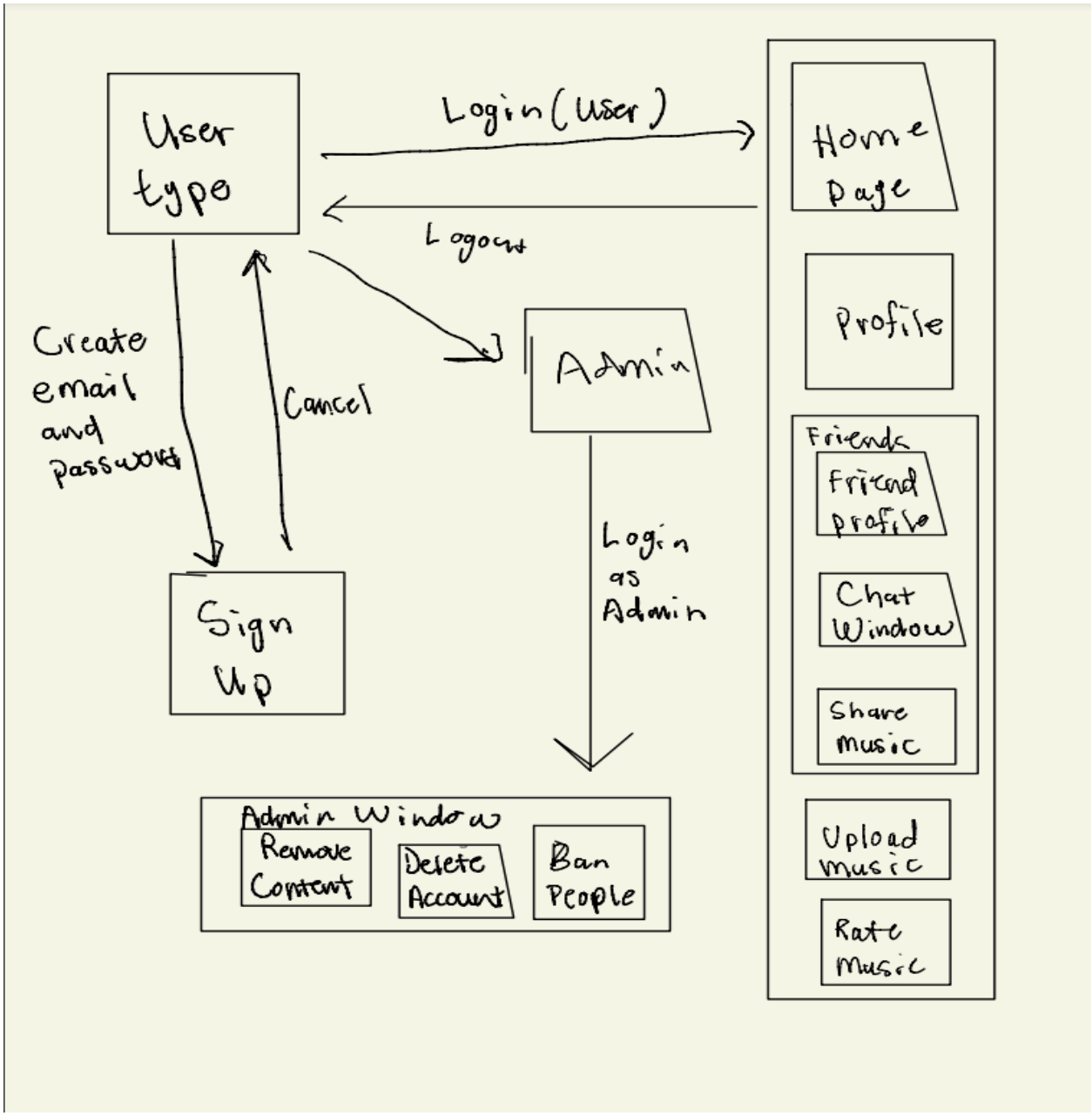
Rating Table:

- **Purpose:** Tracks user ratings for albums and songs.
- **Functionality:** Calculates average ratings and allows users to update their reviews.

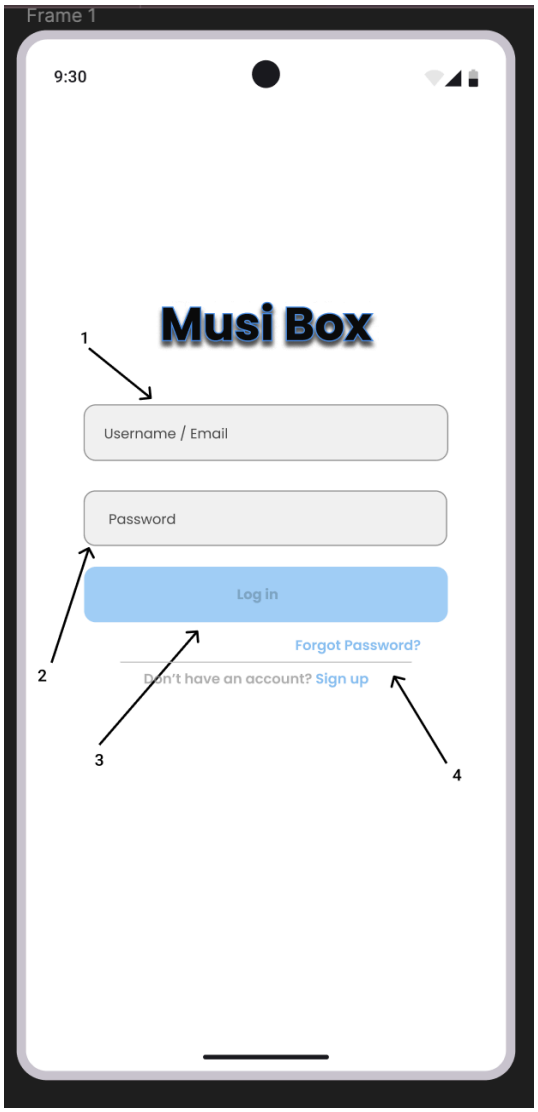
Name : Backend Screen Flow Sketch

Database Tables : User, Admin, Profile, Friend, Chat, Rating, Songs

Developer : Tu Reh



Pages



Name : Login Screen

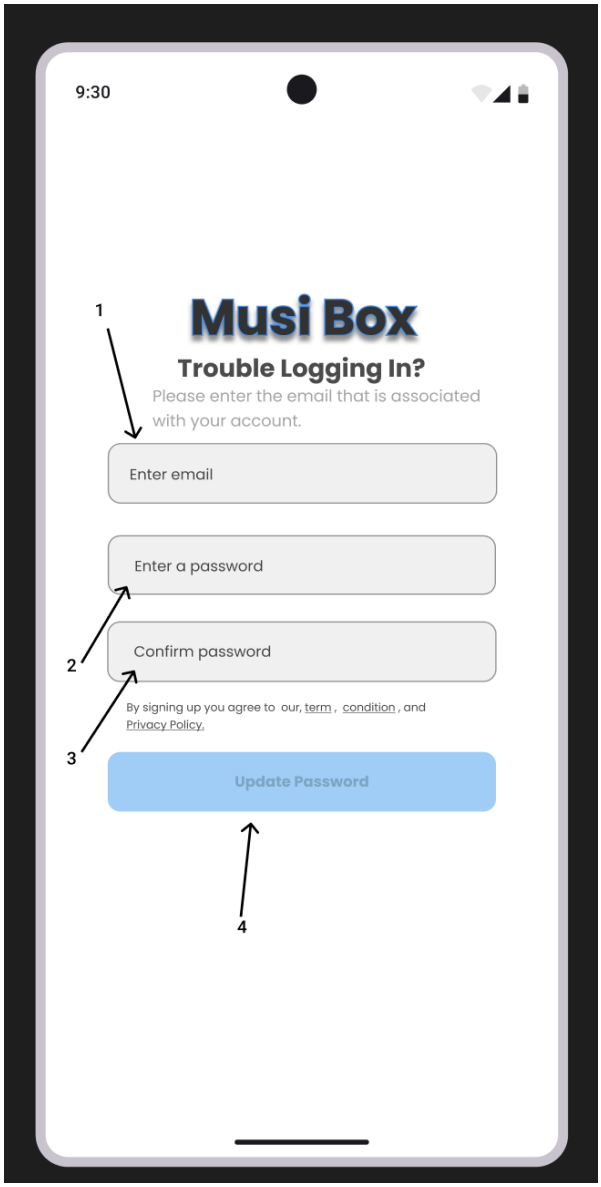
Purpose : Screen for users to login and redirect them to other screens like forget password or sign up.

Developed by: Bae Meh

Interface:

1) **Username Field -** A text input field where users enter their username.

2) **Password Field -** A password input field where users enter their password (masked by default).
3) **Login Button -** A button labeled “Login” that the user clicks to submit their credentials.
4) **Create Account Link -** A clickable link or button labeled “Create Account” that redirects the user to the SignUp screen if they don’t have an account.
5) **Forgot Password Link -** A clickable link labeled “Forgot Password” that redirects the user to a password recovery process if their username and password don’t match or if they’ve forgotten their credentials.



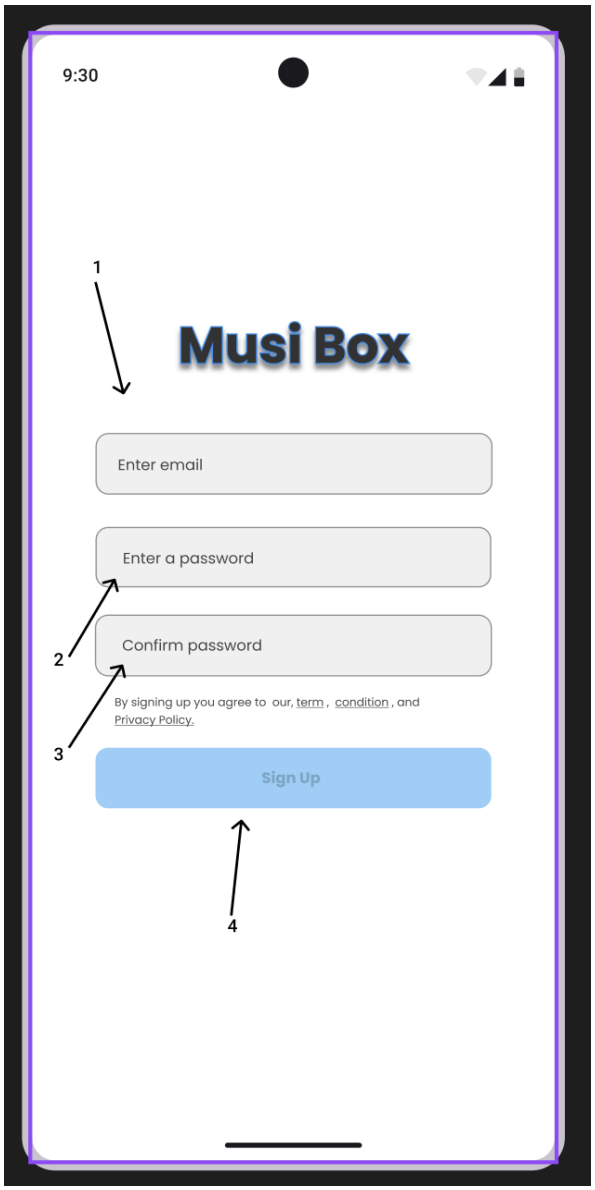
Name: Forget Password Screen

Purpose: A screen for users to reset their password if they forget by verifying their email. Then they have to retype the password twice to make sure they match.

Developer: Bae Meh

Interface:

1) **Email Input Field -** Users can input the email associated with their account.
2) **Password -** A field for users to input the new password they want for their account
3) **Confirm Password -** A field for users to confirm the password that they just entered
4) **Update Password Button -** Button that sends the update request to update their password.



Name: SignUp Screen

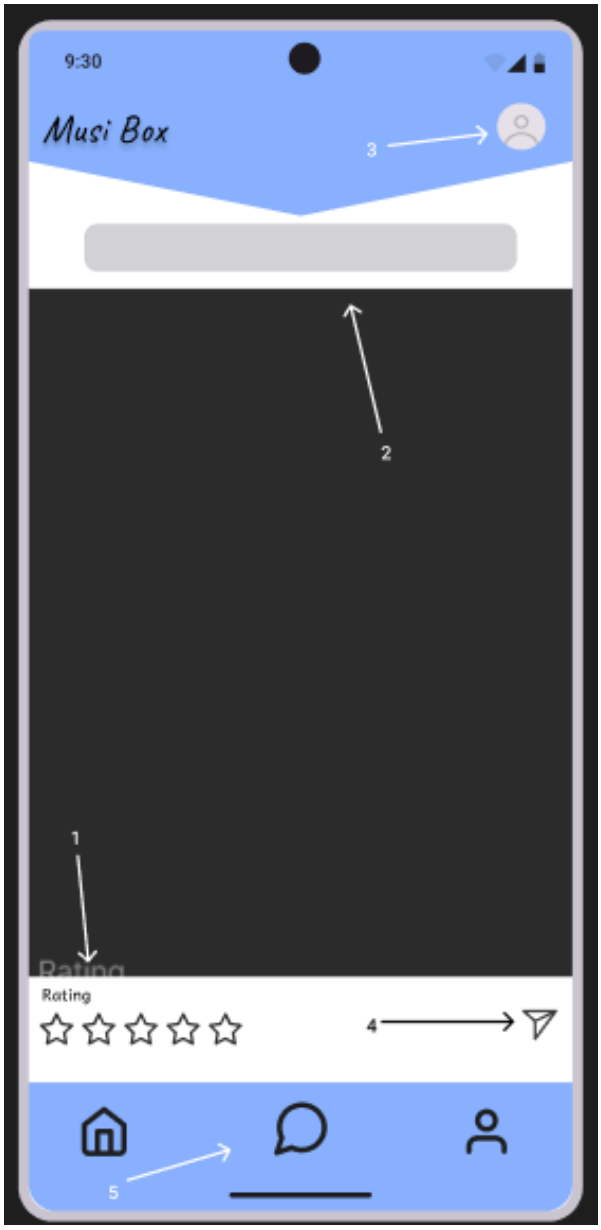
Developer: Ping Wu

Purpose: A page where users can signup and create an account.

Interface:

1) **Email Address Field -** A text input field where users enter their email address.
2) **Password Field -** A password input field where users create a password (masked by default).

3) **Confirm Password Field -** A second password input field where users re-enter the password to confirm it matches the first entry.
4) **Sign Up Button -** A prominently displayed button labeled “Sign Up” that users click to create their account.



Name: Home Page

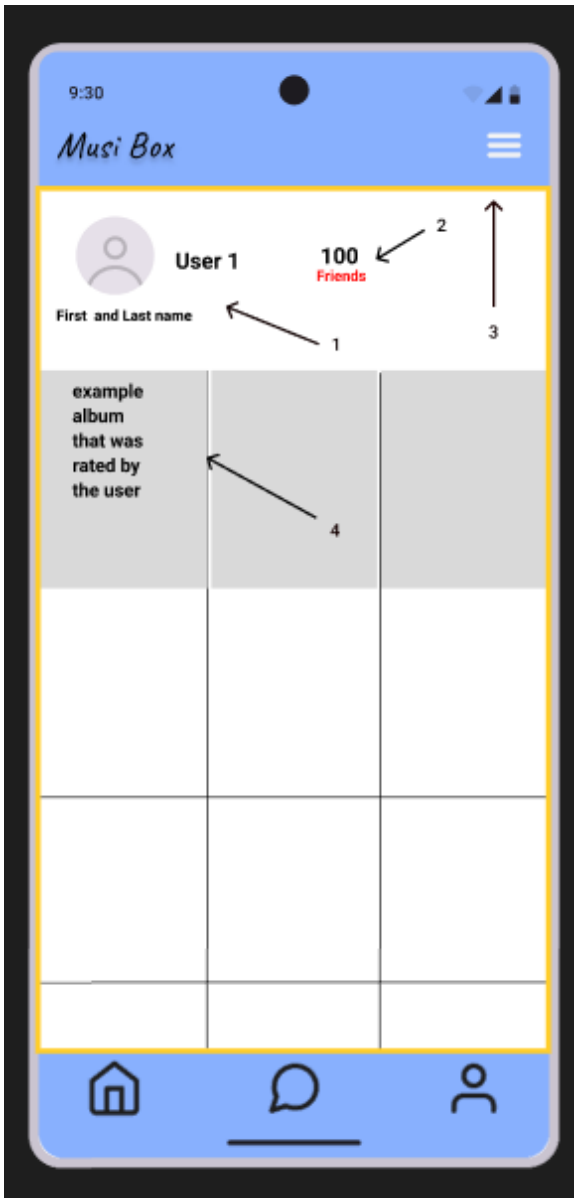
Purpose: A main page where users can see all the songs that are available to rate with functions

like search bar, rating, home bar and more.

Developer: Ping Wu

Interface:

1) **Rating Bar -** A rating input that allows users to rate the song.
2) **Search Bar -** A search input field at the top of the screen allowing users to quickly search for content, users, or items.
3) **Profile image -** The user's profile pic that is also clickable for users to view their profile if clicked.
4) **Share button -** A button for users to share the song with a friend.
5) **Bottom Navigation Menu (On the bottom of every page) -** A fixed navigation bar at the bottom of the screen for easy access to different sections:
    - **Profile: Redirects users to their personal profile.**
    - **Ratings: Allows users to view or give ratings.**
    - **Messages: Takes users to their messages or inbox.**



Name: Profile Page

Purpose: A page for users to see their profile and edit how they want their profile page to look.

Developer: Ping Wu

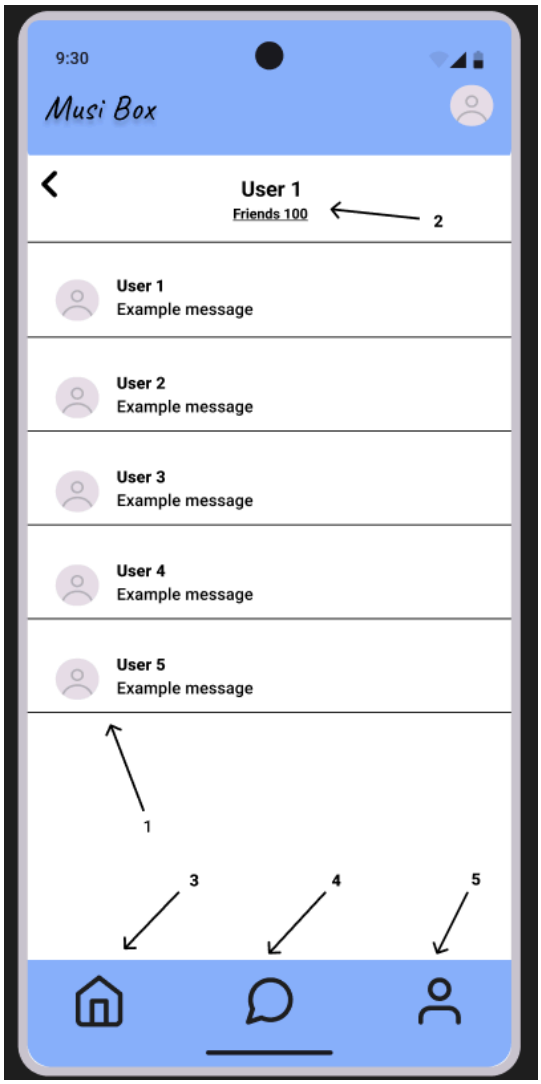
Interface:

1) User Information Section - Displays user-specific details such as the user's name, profile picture, and a brief bio.

2) User's Friend - Displays the user's number of friends..

3) Edit Profile Button - A prominent button that opens a form for users to update their

4) User Activity Section - Showcases recent ratings, achievements, and other user activities to keep users engaged.



Name: Friends Page

Developer: Tu Reh

Purpose: This will display the friends that the user has and how many total people are on their friend list. From this page you will also be able to access friend information, as well as freely change to home page, chat page, and user profile page.

Interface:

1) Friends List Display

• A visually appealing grid or list showcasing the user's current friends, making it easy to browse. This will retrieve the data of every friends the user have including the friends profile picture and name that are displayed for quick identification

2) User Info

• Displays the user name and profile picture and amount of friends they hold.

3) Homepage Button

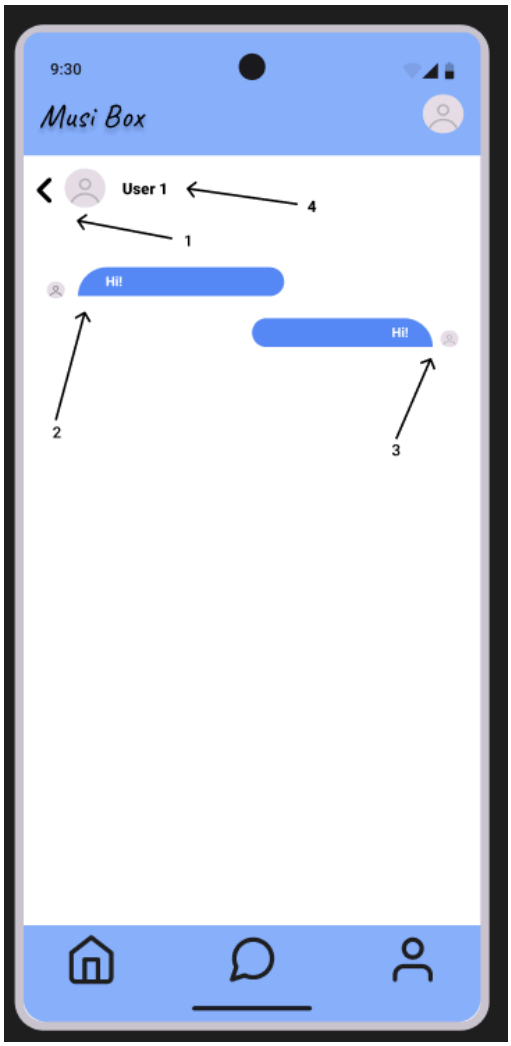
• A button that takes you to the homepage

4) Chat Button

• A button that takes you to the chat page

3) Profile Page Button

• A button that takes you to the profile page button



Name: Message Page

Developer: Tu Reh

Purpose: This page will display the messages between two users. You will be able to see message history and from which message was sent by what user will be indicated by the left to right text design. You will be able to exit from the text chat as well.

Interface:

1) Leave Chat

• A button to leave and back out from a chat server.

2) Receiver Chat

• This will display the text history and text sent by the receiver (friend) of the user as it is

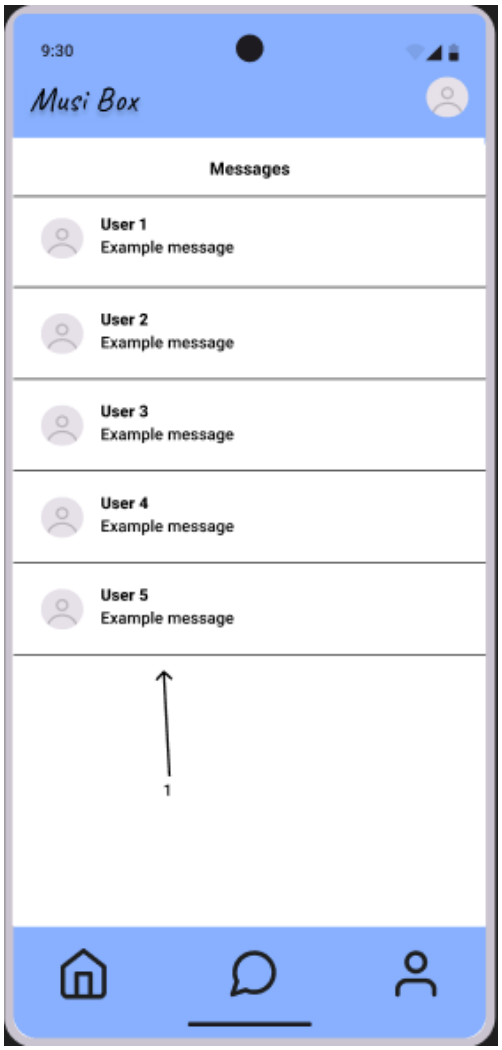
on the left side of the chat server.

3) Sender Chat (User)

• This will display the text history and text sent by the user as it is displayed on the right side of the chat server.

4) Receiver info

• This will display the friend or the receiver info, as it will show their name and profile picture of the person you are chatting with.



Name: Inbox Page

Developer: Haris Khan

Purpose: Page that displays the current chats that the user has

Interface:

1) Message List Display

• A list of recent messages showing sender names, snippets of content, and timestamps for quick reference.

• Clickable Conversations - users can easily navigate to specific conversations by

clicking on the message entries.

• Filtering Options - features filters to sort messages by unread, starred, or important messages for better organization.

• Notification Indicators - visual cues for unread messages to ensure users stay updated on new communications.

Name: Search Friends Page

Developer: Haris Khan

Purpose: A page for users to search for users and add them to their friends list

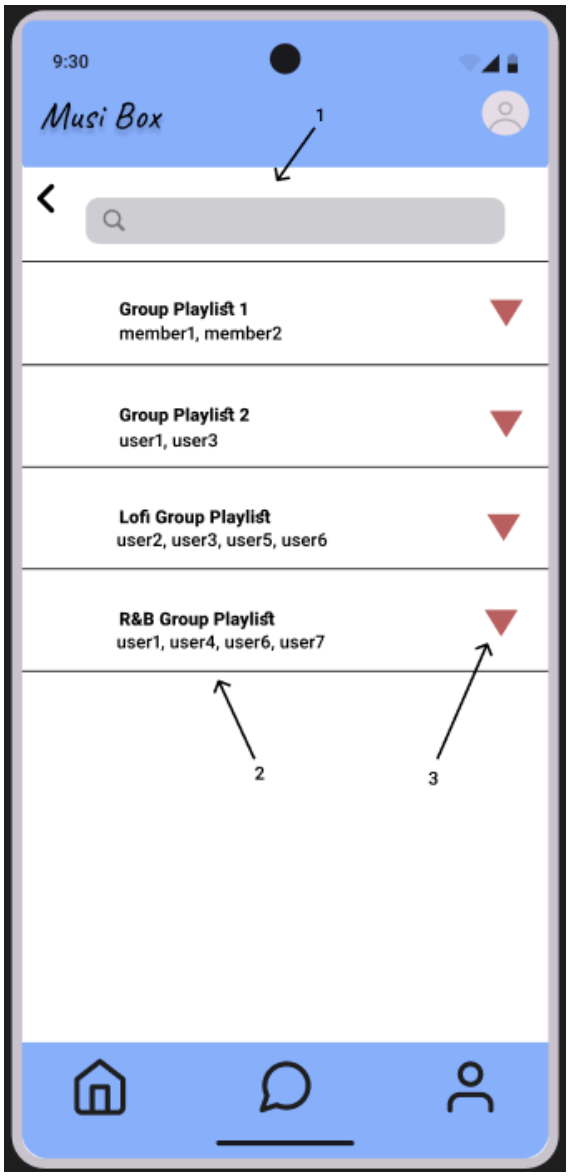
Interface:

1) Search Bar

• A prominent input field at the top for users to enter names or keywords to find specific individuals.

2) Users List

• Displays a list based on keyword searched. If nothing is entered, there are suggested friends based on mutual connections or common interests. Each entry includes profile pictures, names, and brief status indicators.



Name: Group Playlist Page

Developer: Haris Khan

Purpose: A screen for users' group playlists where they can add others users in a group and remove themselves from groups

Interface:

1) Search Bar

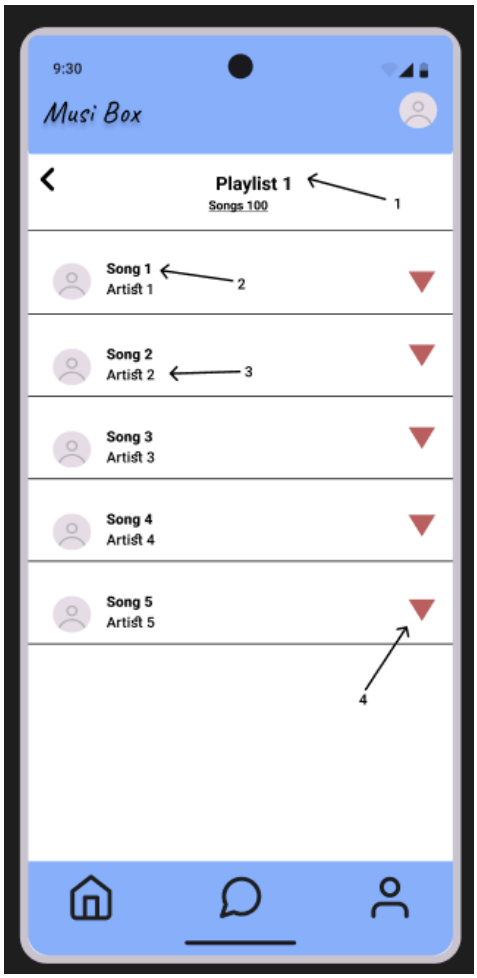
• When creating a playlist the user can add individuals to the playlist they create.

2) Group Playlists

• Lists the playlists that the user is in, displaying the group name and users in the group.

3) Leave Playlist

• User can remove themselves from the group playlist with the drop down option



Name: Playlist with Songs Page

Developer: Tu Reh

Purpose: This page will retrieve the data of songs that have been added to the specific playlist. Users will be able to remove songs from the playlist and also edit the playlist name.

Interface:

1) Playlist Title

• Grabs and display the title of the playlists. You will be able to edit the title

2) Songs title API

• An API holding the data of songs title which will be displayed inside the playlist

3) Artist name API

• An API holding the data of the artists names which will be displayed inside the playlist with the song title

4) Remove Song Button

• This button will allow users to remove the specific song from the playlist which will remove the display of both the song title and artist name